Software Requirement Specification

Project: **Project-3**

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Total SW Requirements	111
Requirement Categories	Functional, Performance, Reliability, Maintainability, Interface, Constraint
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1. Introduction

This Software Requirement Specification (SRS) defines the software-level requirements derived from the System Requirement Document (SRD) for the **Project-3** project, following the IEEE 830 standard. Each requirement is uniquely identified (SW-REQ-XXX), categorized, prioritized, and traced back to its parent system requirement. A total of **111 software requirements** across **6 categories** are documented herein.

2. Software Requirements

2.1 Functional Requirements

SW-REQ-001	**Closed-loop control algorithm**
	The system shall implement a closed-loop control algorithm that uses the potentiometer feedback ADC value to compute actuator drive commands.
	Priority: High **Traces to:** SYS-REQ-001

SW-REQ-011	**Potentiometer feedback configuration**
	The system shall configure a single analog input channel for the actuator potentiometer feedback.
	Priority: High **Traces to:** SYS-REQ-008

SW-REQ-012	**ADC read function**
	The system shall read the actuator feedback potentiometer via the ADC and provide the raw and scaled values to the control algorithm.
	Priority: High **Traces to:** SYS-REQ-009

SW-REQ-013	**ADC scaling to position**
	The system shall scale the ADC reading to a normalized position value in the range 0.0 to 1.0 using configurable minimum and maximum calibration values.
	Priority: Medium **Traces to:** SYS-REQ-009

SW-REQ-014	Actuator control output
	The system shall compute and update actuator control output based on the measured ADC position and the commanded target position at each control cycle.
	Priority: High **Traces to:** SYS-REQ-010

SW-REQ-017	Shipping position threshold configuration
	The system shall store a configurable ADC threshold value representing the shipping position in non-volatile memory and use it for control decisions.
	Priority: High **Traces to:** SYS-REQ-011

SW-REQ-018	Stop at shipping position
	The system shall stop actuator drive within 20 ms after the ADC feedback value crosses the configured shipping position threshold.
	Priority: High **Traces to:** SYS-REQ-011

SW-REQ-022	Continuous ADC acquisition loop
	The system shall continuously sample the actuator potentiometer using the internal ADC without gaps longer than 2 ms during normal operation.
	Priority: High **Traces to:** SYS-REQ-013

SW-REQ-025	ADC as primary feedback variable
	The system shall use the latest ADC sample as the primary feedback variable for all actuator control computations.
	Priority: High **Traces to:** SYS-REQ-016

SW-REQ-026	ADC-only flap control inputs
	The system shall compute flap actuation commands using only the measured ADC feedback value and configured thresholds, with no other sensor inputs.
	Priority: High **Traces to:** SYS-REQ-017

SW-REQ-028	Control decisions based on real-time ADC
	The system shall base each control decision on the most recent ADC sample that is no older than 2 ms.
	Priority: **High** **Traces to:** SYS-REQ-019

SW-REQ-030	ADC validity monitoring
	The system shall flag ADC feedback as invalid if no new sample is acquired within 2 ms or if the ADC value is out of the 0 to 4095 range.
	Priority: **High** **Traces to:** SYS-REQ-020

SW-REQ-033	Disable drive on ADC out of range
	The system shall disable the actuator drive output within one control loop cycle when the measured ADC value is less than 500 or greater than 2600.
	Priority: **High** **Traces to:** SYS-REQ-022

SW-REQ-034	Stop at target ADC
	The system shall stop actuator movement when the measured ADC value is within ±1 count of the configured target ADC value.
	Priority: **High** **Traces to:** SYS-REQ-023

SW-REQ-036	Compute scaled supply voltage
	The system shall compute the external supply voltage by applying the configured divider ratio to the measured ADC value.
	Priority: **High** **Traces to:** SYS-REQ-025

SW-REQ-039	Overvoltage detection
	The system shall declare an overvoltage condition when the computed supply voltage is greater than or equal to Overvoltage_Threshold.
	Priority: **High** **Traces to:** SYS-REQ-027

SW-REQ-040	**Disable drive on overvoltage**
	The system shall disable the actuator drive output within one control loop cycle of detecting an overvoltage condition.
	Priority: High Traces to: SYS-REQ-028

SW-REQ-041	**Block actuator commands during overvoltage**
	The system shall ignore or reject all actuator movement commands while an overvoltage condition is present.
	Priority: High Traces to: SYS-REQ-029

SW-REQ-042	**Enter fault state on overvoltage**
	The system shall transition to the FAULT state within one control loop cycle of detecting an overvoltage condition.
	Priority: High Traces to: SYS-REQ-030

SW-REQ-044	**Maintain FAULT state until clear conditions**
	The system shall remain in FAULT state until both conditions are true: (1) the measured supply voltage is within the acceptable limits, and (2) a system reset or an authorized UART command to clear fault is received.
	Priority: High Traces to: SYS-REQ-032

SW-REQ-046	**Continuous overvoltage monitoring**
	The system shall sample the supply voltage and evaluate overvoltage status at least once every 100 ms during all active system states.
	Priority: High Traces to: SYS-REQ-033

SW-REQ-050	**ACTUATOR_ON execution**
	Upon receiving a valid "ACTUATOR_ON" command, the system shall initiate actuator operation according to the configured control logic within 100 ms.

	Priority: **High** **Traces to:** SYS-REQ-036

SW-REQ-051	**ACTUATOR_OFF execution**
	Upon receiving a valid "ACTUATOR_OFF" command, the system shall disable the actuator output within 50 ms.
	Priority: **High** **Traces to:** SYS-REQ-037

SW-REQ-052	**Ignore invalid UART commands**
	The system shall take no actuator state change when an invalid UART command is received.
	Priority: **Medium** **Traces to:** SYS-REQ-038

SW-REQ-053	**Log invalid UART commands**
	The system shall log each invalid UART command with a timestamp and the received raw command string.
	Priority: **Medium** **Traces to:** SYS-REQ-038

SW-REQ-058	**ADC resolution configuration**
	The system shall configure the ADC to operate at 12-bit resolution.
	Priority: **High** **Traces to:** SYS-REQ-043

SW-REQ-064	**PWM timer initialization**
	The system shall initialize an advanced PWM timer to generate actuator control signals.
	Priority: **High** **Traces to:** SYS-REQ-047

SW-REQ-070	**State machine implementation**
	The system shall implement a state machine with the states INIT, IDLE, ACTIVE, SHIPPING_LOCK, and FAULT.
	Priority: **High** **Traces to:** SYS-REQ-050

SW-REQ-071	Initial state on reset
	The system shall enter the INIT state on every reset or power-up.
	Priority: High **Traces to:** SYS-REQ-050

SW-REQ-072	Nominal state transitions
	The system shall transition from INIT to IDLE after successful initialization, from IDLE to ACTIVE upon receiving an activate command, from ACTIVE to SHIPPING_LOCK upon receiving a shipping lock command, and from SHIPPING_LOCK to IDLE upon receiving an unlock command.
	Priority: Medium **Traces to:** SYS-REQ-050

SW-REQ-077	ADC-only feedback for control
	The system shall compute closed-loop control outputs using only raw ADC feedback values and internal control state, without using any other sensor inputs.
	Priority: High **Traces to:** SYS-REQ-053

SW-REQ-079	Threshold-based actuator stop
	The system shall command the actuator to stop within one control loop cycle when the ADC value is at or beyond the configured stop thresholds.
	Priority: High **Traces to:** SYS-REQ-055

SW-REQ-086	Overcurrent shutdown
	The system shall monitor motor current and, when the measured current exceeds a configurable overcurrent threshold, disable the actuator output and raise a fault.
	Priority: High **Traces to:** SYS-REQ-062

SW-REQ-087	Thermal shutdown
	The system shall monitor the MCU internal temperature sensor and, when the measured temperature exceeds a configurable thermal threshold, disable the actuator output and raise a fault.

	Priority: **High** **Traces to:** SYS-REQ-062

SW-REQ-088	**Stall detection**
	The system shall detect a stall condition when the actuator is commanded to move and the ADC feedback change remains below a configurable delta for a configurable duration, and shall raise a fault.
	Priority: **High** **Traces to:** SYS-REQ-062

SW-REQ-089	**Short-circuit protection**
	The system shall detect a short-circuit condition using available hardware indicators or current measurements and, upon detection, disable the actuator output and raise a fault.
	Priority: **High** **Traces to:** SYS-REQ-062

SW-REQ-090	**Undervoltage detection**
	The system shall monitor supply voltage and, when the measured voltage is below 9 V, disable the actuator output and raise a fault.
	Priority: **High** **Traces to:** SYS-REQ-062

SW-REQ-092	**Fault state transition**
	The system shall transition the state machine to FAULT state in the same processing cycle in which any fault condition is detected.
	Priority: **High** **Traces to:** SYS-REQ-063

SW-REQ-096	**Continuous ADC acquisition**
	The system shall sample the ADC continuously during powered operation at a fixed periodic rate and make the latest sample available to the control loop each cycle.
	Priority: **High** **Traces to:** SYS-REQ-067

SW-REQ-097	**ADC feedback control**

	The system shall compute and update actuator commands using the most recent ADC feedback on each control loop iteration. **Priority: High Traces to:** SYS-REQ-068

SW-REQ-098	**Hard stop at Shipping_ADC_Value**
	The system shall stop actuator motion and prevent further motion toward the limit when the ADC feedback reaches the configured Shipping_ADC_Value threshold. **Priority: High Traces to:** SYS-REQ-069

SW-REQ-100	**Fault condition detection**
	The system shall monitor all defined fault inputs and detect a fault within 100 ms of the fault condition becoming true. **Priority: High Traces to:** SYS-REQ-071

SW-REQ-101	**Fault reporting**
	The system shall log each detected fault with a timestamp and fault code to non-volatile storage within 1 s of detection. **Priority: High Traces to:** SYS-REQ-071

SW-REQ-102	**State transition on fault**
	The system shall transition from NORMAL to FAULT state within 500 ms after any fault is detected. **Priority: High Traces to:** SYS-REQ-071

SW-REQ-103	**State transition recovery**
	The system shall transition from FAULT to NORMAL state only after all faults are cleared and a recovery command is received. **Priority: Medium Traces to:** SYS-REQ-071

2.2 Performance Requirements

SW-REQ-002	ADC sampling for feedback
	The system shall sample the feedback ADC channel at a configurable rate between 100 Hz and 1 kHz with a default of 500 Hz.
	Priority: Medium **Traces to:** SYS-REQ-001

SW-REQ-015	Control loop timing
	The system shall execute the control loop at a fixed period of 10 ms ±1 ms.
	Priority: Medium **Traces to:** SYS-REQ-010

SW-REQ-023	ADC resolution configuration
	The system shall configure the internal ADC to operate at 12-bit resolution or higher.
	Priority: High **Traces to:** SYS-REQ-014

SW-REQ-024	ADC sampling rate
	The system shall sample the internal ADC at a rate of at least 1000 samples per second.
	Priority: High **Traces to:** SYS-REQ-015

SW-REQ-037	Continuous supply monitoring
	The system shall sample and update the computed supply voltage at least once per control loop cycle.
	Priority: High **Traces to:** SYS-REQ-026

SW-REQ-045	FAULT exit voltage check
	The system shall evaluate the supply voltage for FAULT exit using the configured acceptable limits at least once every 100 ms while in FAULT state.
	Priority: Medium **Traces to:** SYS-REQ-032

SW-REQ-060	ADC sampling frequency

	The system shall sample the ADC channel at a rate of at least 1 kHz.
	Priority: High Traces to: SYS-REQ-045

SW-REQ-066	**DMA usage for data transfer**
	The system shall configure DMA support for ADC data transfers to reduce CPU load during sampling.
	Priority: Medium Traces to: SYS-REQ-047

SW-REQ-068	**Control loop timing stability**
	The system shall configure the system clock and timers such that the control loop period is 1 ms ±1%.
	Priority: High Traces to: SYS-REQ-048

SW-REQ-078	**Control loop period**
	The system shall execute the control loop at a fixed period of 1.0 ms or less (≥ 1 kHz).
	Priority: High Traces to: SYS-REQ-054

SW-REQ-080	**Anti-oscillation hysteresis**
	The system shall apply a configurable hysteresis band around stop thresholds to prevent rapid toggling of actuator commands.
	Priority: Medium Traces to: SYS-REQ-056

SW-REQ-085	**UART transmit interval**
	The system shall transmit UART messages at a periodic interval less than or equal to 100 ms during normal operation.
	Priority: High Traces to: SYS-REQ-061

2.3 Reliability Requirements

SW-REQ-007	Transient immunity robustness
	The system shall implement input validation and fault state handling for ADC and UART data to support ISO 7637 transient immunity testing.
	Priority: Medium **Traces to:** SYS-REQ-004

SW-REQ-008	EMC immunity robustness
	The system shall detect and reject out-of-range ADC readings outside the calibrated potentiometer limits and enter a safe state.
	Priority: High **Traces to:** SYS-REQ-005

SW-REQ-016	Safe state on feedback fault
	The system shall disable actuator drive and report a fault if the feedback ADC value is invalid for more than 50 ms.
	Priority: High **Traces to:** SYS-REQ-010

SW-REQ-056	Safe outputs on startup
	The system shall keep all actuator control outputs (PA5 GPIO and PWM outputs) disabled or in a high-impedance state until initialization completes and a 100 ms power stabilization delay has elapsed after reset.
	Priority: High **Traces to:** SYS-REQ-041

SW-REQ-057	Startup delay for transient suppression
	The system shall delay enabling the control loop and actuator outputs for at least 50 ms after reset to avoid reacting to power-supply transients.
	Priority: Medium **Traces to:** SYS-REQ-042

SW-REQ-061	ADC overvoltage detection
	The system shall detect ADC input saturation by identifying 5 consecutive samples at the maximum ADC count and shall transition to the FAULT state when saturation is detected.

	Priority: **High** **Traces to:** SYS-REQ-046

SW-REQ-067	**Independent watchdog service**
	The system shall enable the independent watchdog and refresh it at least once within 50% of the watchdog timeout period during normal operation.
	Priority: **High** **Traces to:** SYS-REQ-047

SW-REQ-073	**Fault state transitions**
	The system shall transition to the FAULT state from any state when a fault condition is detected and shall remain in FAULT until a reset occurs.
	Priority: **High** **Traces to:** SYS-REQ-050

SW-REQ-074	**Watchdog initialization**
	The system shall initialize an independent hardware watchdog timer during startup and configure it with the platform's supported timeout value.
	Priority: **High** **Traces to:** SYS-REQ-051

SW-REQ-075	**Watchdog service interval**
	The system shall service the independent watchdog at least once within each watchdog timeout period during normal operation.
	Priority: **High** **Traces to:** SYS-REQ-051

SW-REQ-076	**Reset on detected malfunction**
	The system shall intentionally stop servicing the watchdog to trigger a reset when a fatal software malfunction is detected.
	Priority: **High** **Traces to:** SYS-REQ-052

SW-REQ-091	**Watchdog supervision**
	The system shall enable the hardware watchdog, service it periodically during normal execution, and on watchdog timeout shall reset the system and record a watchdog fault indication.

	Priority: **High** **Traces to:** SYS-REQ-062

2.4 Maintainability Requirements

SW-REQ-009	**MISRA-C compliance**
	The system shall be implemented in C and adhere to MISRA-C:2012 guidelines with documented deviations.
	Priority: **Medium** **Traces to:** SYS-REQ-006

SW-REQ-038	**Configurable overvoltage threshold parameter**
	The system shall store an Overvoltage_Threshold parameter in non-volatile memory and make it available to the voltage monitoring logic on startup.
	Priority: **Medium** **Traces to:** SYS-REQ-027

SW-REQ-069	**Layered architecture enforcement**
	The system shall structure software into HAL, Driver, Middleware, and Application layers, and shall prevent direct hardware access from the Application layer by requiring all hardware interactions to pass through lower layers.
	Priority: **Medium** **Traces to:** SYS-REQ-049

2.5 Interface Requirements

SW-REQ-003	**UART command interface**
	The system shall accept control commands over UART to set target actuator position and control mode.
	Priority: **High** **Traces to:** SYS-REQ-002

SW-REQ-004	**UART diagnostics reporting**

	The system shall provide diagnostic messages over UART including current ADC value, target position, control output, and fault status on request. **Priority:** Medium **Traces to:** SYS-REQ-002

SW-REQ-005	**UART protocol settings**
	The system shall support UART communication at 115200 baud, 8 data bits, no parity, and 1 stop bit. **Priority:** Medium **Traces to:** SYS-REQ-002

SW-REQ-019	**UART ON command handling**
	The system shall accept a UART command to enable actuator drive and transition to an ON state within 50 ms of command receipt. **Priority:** High **Traces to:** SYS-REQ-012

SW-REQ-020	**UART OFF command handling**
	The system shall accept a UART command to disable actuator drive and transition to an OFF state within 50 ms of command receipt. **Priority:** High **Traces to:** SYS-REQ-012

SW-REQ-021	**UART command acknowledgement**
	The system shall transmit a UART acknowledgement message for each valid ON or OFF command within 20 ms of processing the command. **Priority:** Medium **Traces to:** SYS-REQ-012

SW-REQ-035	**Configure supply voltage ADC channel**
	The system shall configure a dedicated ADC channel to sample the external supply voltage through a resistive divider network. **Priority:** High **Traces to:** SYS-REQ-024

SW-REQ-043	**Overvoltage UART fault message**

	The system shall transmit the UART diagnostic message "OVERVOLTAGE_FAULT_ACTIVE" within 100 ms when the measured supply voltage is greater than or equal to the configured Overvoltage_Threshold.
	Priority: **High** **Traces to:** SYS-REQ-031

SW-REQ-047	**UART actuator control interface**
	The system shall accept actuator control commands over the UART interface when the UART link is active.
	Priority: **High** **Traces to:** SYS-REQ-034

SW-REQ-048	**Support ACTUATOR_ON command**
	The system shall recognize the UART command "ACTUATOR_ON" as a valid command.
	Priority: **High** **Traces to:** SYS-REQ-035

SW-REQ-049	**Support ACTUATOR_OFF command**
	The system shall recognize the UART command "ACTUATOR_OFF" as a valid command.
	Priority: **High** **Traces to:** SYS-REQ-035

SW-REQ-062	**ADC pin initialization**
	The system shall initialize the ADC input on pin PA1 for sensor acquisition.
	Priority: **High** **Traces to:** SYS-REQ-047

SW-REQ-063	**Actuator GPIO initialization**
	The system shall configure pin PA5 as a GPIO output for actuator control.
	Priority: **High** **Traces to:** SYS-REQ-047

SW-REQ-065	**UART initialization**
	The system shall initialize at least one UART peripheral for external communication.

	Priority: Medium **Traces to:** SYS-REQ-047

SW-REQ-082	**Periodic UART status message**
	The system shall transmit a UART message at a configurable periodic interval containing raw ADC value, system state, shipping lock status, and fault flags.
	Priority: Medium **Traces to:** SYS-REQ-058

SW-REQ-083	**UART baud rate configuration**
	The system shall configure the UART interface to 115200 baud.
	Priority: High **Traces to:** SYS-REQ-059

SW-REQ-084	**UART frame format configuration**
	The system shall configure the UART interface to 8 data bits, no parity, and 1 stop bit (8N1).
	Priority: High **Traces to:** SYS-REQ-060

SW-REQ-099	**UART ON/OFF control**
	The system shall accept UART commands to turn the actuator ON and OFF and shall update the actuator enable state accordingly.
	Priority: High **Traces to:** SYS-REQ-070

SW-REQ-104	**State transition verification output**
	The system shall provide a software-accessible status indicator that reflects the current system state and last transition cause.
	Priority: Medium **Traces to:** SYS-REQ-071

2.6 Constraint Requirements

SW-REQ-006	**Environmental condition handling**

	The system shall provide software configuration options to adjust ADC sampling rate, control gains, and UART timeouts to support ISO 16750 environmental conditions testing. **Priority:** Low **Traces to:** SYS-REQ-003

SW-REQ-010	**12V supply assumption**
	The system shall assume a 12V actuator supply and use this assumption for scaling any control output duty cycle or drive limits. **Priority:** Medium **Traces to:** SYS-REQ-007

SW-REQ-027	**Flap actuation range enforcement**
	The system shall permit flap actuation only when the ADC feedback value is within 500 to 2600 counts, and shall stop actuation when the value is outside this range. **Priority:** High **Traces to:** SYS-REQ-018

SW-REQ-029	**Prohibit open-loop control**
	The system shall inhibit actuator drive if valid ADC feedback is unavailable, thereby preventing open-loop control under any operating condition. **Priority:** High **Traces to:** SYS-REQ-020

SW-REQ-031	**Permit motion only in ADC window**
	The system shall permit actuator drive commands only when the measured flap position ADC value is within 500 to 2600 inclusive. **Priority:** High **Traces to:** SYS-REQ-021

SW-REQ-032	**Reject out-of-window targets**
	The system shall reject any configured target ADC value that is outside the range 500 to 2600 inclusive. **Priority:** High **Traces to:** SYS-REQ-021

SW-REQ-054	**Nominal voltage configuration**

	The system shall store the nominal supply voltage value as 12 V for use in diagnostics and reporting. **Priority:** Low **Traces to:** SYS-REQ-039

SW-REQ-055	**Operating voltage range configuration**
	The system shall configure the acceptable supply voltage limits to 9 V minimum and 16 V maximum for operating range checks. **Priority:** High **Traces to:** SYS-REQ-040

SW-REQ-059	**ADC reference voltage configuration**
	The system shall configure the ADC reference voltage to 3.3 V. **Priority:** High **Traces to:** SYS-REQ-044

SW-REQ-081	**Raw ADC usage only**
	The system shall not perform any conversion of ADC values into physical position or angle units for control or decision logic. **Priority:** High **Traces to:** SYS-REQ-057

SW-REQ-093	**Operating temperature range handling**
	The system shall correctly process temperature sensor inputs across the range of -40°C to +85°C without overflow or loss of resolution. **Priority:** Medium **Traces to:** SYS-REQ-064

SW-REQ-094	**Storage temperature range handling**
	The system shall preserve configuration and calibration data in non-volatile memory without software-enforced invalidation when the device experiences storage temperatures from -40°C to +105°C. **Priority:** Medium **Traces to:** SYS-REQ-065

SW-REQ-095	**Humidity constraint compliance**

	The system shall not depend on humidity measurements for operation and shall not execute any condensation-detection logic, assuming a non-condensing environment up to 95% RH. **Priority:** Low **Traces to:** SYS-REQ-066

SW-REQ-105	**Architecture diagram artifact generation**
	The system shall generate an exportable system architecture diagram file in PDF format from the project documentation repository. **Priority:** Low **Traces to:** SYS-REQ-072

SW-REQ-106	**Schematic artifact generation**
	The system shall include a software-controlled export of the schematic in PDF format as part of the release package. **Priority:** Low **Traces to:** SYS-REQ-073

SW-REQ-107	**PCB layout artifact generation**
	The system shall include a software-controlled export of PCB layout files in Gerber format within the release package. **Priority:** Low **Traces to:** SYS-REQ-074

SW-REQ-108	**Firmware source code packaging**
	The system shall provide the complete firmware source code in a version-controlled repository and include a tagged release for delivery. **Priority:** High **Traces to:** SYS-REQ-075

SW-REQ-109	**Calibration procedure documentation**
	The system shall include a software-accessible calibration procedure document in PDF format within the release package. **Priority:** Medium **Traces to:** SYS-REQ-076

SW-REQ-110	**Test report generation**

	The system shall generate a test report in PDF format summarizing test cases, results, and pass/fail status for each build.
	Priority: **Medium** **Traces to:** SYS-REQ-077

SW-REQ-111	**Demonstration video packaging**
	The system shall include a demonstration video file in MP4 format within the release package.
	Priority: **Low** **Traces to:** SYS-REQ-078

3. Traceability Matrix

The table below maps each Software Requirement (SW-REQ) back to its parent System Requirement (SYS-REQ) for full traceability.

SW-REQ ID	Title	SYS-REQ ID	Priority
SW-REQ-001	Closed-loop control algorithm	SYS-REQ-001	High
SW-REQ-002	ADC sampling for feedback	SYS-REQ-001	Medium
SW-REQ-003	UART command interface	SYS-REQ-002	High
SW-REQ-004	UART diagnostics reporting	SYS-REQ-002	Medium
SW-REQ-005	UART protocol settings	SYS-REQ-002	Medium
SW-REQ-006	Environmental condition handling	SYS-REQ-003	Low
SW-REQ-007	Transient immunity robustness	SYS-REQ-004	Medium
SW-REQ-008	EMC immunity robustness	SYS-REQ-005	High
SW-REQ-009	MISRA-C compliance	SYS-REQ-006	Medium
SW-REQ-010	12V supply assumption	SYS-REQ-007	Medium
SW-REQ-011	Potentiometer feedback configuration	SYS-REQ-008	High
SW-REQ-012	ADC read function	SYS-REQ-009	High
SW-REQ-013	ADC scaling to position	SYS-REQ-009	Medium
SW-REQ-014	Actuator control output	SYS-REQ-010	High
SW-REQ-015	Control loop timing	SYS-REQ-010	Medium
SW-REQ-016	Safe state on feedback fault	SYS-REQ-010	High
SW-REQ-017	Shipping position threshold configuration	SYS-REQ-011	High
SW-REQ-018	Stop at shipping position	SYS-REQ-011	High
SW-REQ-019	UART ON command handling	SYS-REQ-012	High

SW-REQ-020	UART OFF command handling	SYS-REQ-012	High
SW-REQ-021	UART command acknowledgement	SYS-REQ-012	Medium
SW-REQ-022	Continuous ADC acquisition loop	SYS-REQ-013	High
SW-REQ-023	ADC resolution configuration	SYS-REQ-014	High
SW-REQ-024	ADC sampling rate	SYS-REQ-015	High
SW-REQ-025	ADC as primary feedback variable	SYS-REQ-016	High
SW-REQ-026	ADC-only flap control inputs	SYS-REQ-017	High
SW-REQ-027	Flap actuation range enforcement	SYS-REQ-018	High
SW-REQ-028	Control decisions based on real-time ADC	SYS-REQ-019	High
SW-REQ-029	Prohibit open-loop control	SYS-REQ-020	High
SW-REQ-030	ADC validity monitoring	SYS-REQ-020	High
SW-REQ-031	Permit motion only in ADC window	SYS-REQ-021	High
SW-REQ-032	Reject out-of-window targets	SYS-REQ-021	High
SW-REQ-033	Disable drive on ADC out of range	SYS-REQ-022	High
SW-REQ-034	Stop at target ADC	SYS-REQ-023	High
SW-REQ-035	Configure supply voltage ADC channel	SYS-REQ-024	High
SW-REQ-036	Compute scaled supply voltage	SYS-REQ-025	High
SW-REQ-037	Continuous supply monitoring	SYS-REQ-026	High
SW-REQ-038	Configurable overvoltage threshold parameter	SYS-REQ-027	Medium
SW-REQ-039	Overvoltage detection	SYS-REQ-027	High
SW-REQ-040	Disable drive on overvoltage	SYS-REQ-028	High
SW-REQ-041	Block actuator commands during overvoltage	SYS-REQ-029	High
SW-REQ-042	Enter fault state on overvoltage	SYS-REQ-030	High

SW-REQ-043	Overvoltage UART fault message	SYS-REQ-031	High
SW-REQ-044	Maintain FAULT state until clear conditions	SYS-REQ-032	High
SW-REQ-045	FAULT exit voltage check	SYS-REQ-032	Medium
SW-REQ-046	Continuous overvoltage monitoring	SYS-REQ-033	High
SW-REQ-047	UART actuator control interface	SYS-REQ-034	High
SW-REQ-048	Support ACTUATOR_ON command	SYS-REQ-035	High
SW-REQ-049	Support ACTUATOR_OFF command	SYS-REQ-035	High
SW-REQ-050	ACTUATOR_ON execution	SYS-REQ-036	High
SW-REQ-051	ACTUATOR_OFF execution	SYS-REQ-037	High
SW-REQ-052	Ignore invalid UART commands	SYS-REQ-038	Medium
SW-REQ-053	Log invalid UART commands	SYS-REQ-038	Medium
SW-REQ-054	Nominal voltage configuration	SYS-REQ-039	Low
SW-REQ-055	Operating voltage range configuration	SYS-REQ-040	High
SW-REQ-056	Safe outputs on startup	SYS-REQ-041	High
SW-REQ-057	Startup delay for transient suppression	SYS-REQ-042	Medium
SW-REQ-058	ADC resolution configuration	SYS-REQ-043	High
SW-REQ-059	ADC reference voltage configuration	SYS-REQ-044	High
SW-REQ-060	ADC sampling frequency	SYS-REQ-045	High
SW-REQ-061	ADC overvoltage detection	SYS-REQ-046	High
SW-REQ-062	ADC pin initialization	SYS-REQ-047	High
SW-REQ-063	Actuator GPIO initialization	SYS-REQ-047	High
SW-REQ-064	PWM timer initialization	SYS-REQ-047	High
SW-REQ-065	UART initialization	SYS-REQ-047	Medium

SW-REQ-066	DMA usage for data transfer	SYS-REQ-047	Medium
SW-REQ-067	Independent watchdog service	SYS-REQ-047	High
SW-REQ-068	Control loop timing stability	SYS-REQ-048	High
SW-REQ-069	Layered architecture enforcement	SYS-REQ-049	Medium
SW-REQ-070	State machine implementation	SYS-REQ-050	High
SW-REQ-071	Initial state on reset	SYS-REQ-050	High
SW-REQ-072	Nominal state transitions	SYS-REQ-050	Medium
SW-REQ-073	Fault state transitions	SYS-REQ-050	High
SW-REQ-074	Watchdog initialization	SYS-REQ-051	High
SW-REQ-075	Watchdog service interval	SYS-REQ-051	High
SW-REQ-076	Reset on detected malfunction	SYS-REQ-052	High
SW-REQ-077	ADC-only feedback for control	SYS-REQ-053	High
SW-REQ-078	Control loop period	SYS-REQ-054	High
SW-REQ-079	Threshold-based actuator stop	SYS-REQ-055	High
SW-REQ-080	Anti-oscillation hysteresis	SYS-REQ-056	Medium
SW-REQ-081	Raw ADC usage only	SYS-REQ-057	High
SW-REQ-082	Periodic UART status message	SYS-REQ-058	Medium
SW-REQ-083	UART baud rate configuration	SYS-REQ-059	High
SW-REQ-084	UART frame format configuration	SYS-REQ-060	High
SW-REQ-085	UART transmit interval	SYS-REQ-061	High
SW-REQ-086	Overcurrent shutdown	SYS-REQ-062	High
SW-REQ-087	Thermal shutdown	SYS-REQ-062	High
SW-REQ-088	Stall detection	SYS-REQ-062	High

SW-REQ-089	Short-circuit protection	SYS-REQ-062	High
SW-REQ-090	Undervoltage detection	SYS-REQ-062	High
SW-REQ-091	Watchdog supervision	SYS-REQ-062	High
SW-REQ-092	Fault state transition	SYS-REQ-063	High
SW-REQ-093	Operating temperature range handling	SYS-REQ-064	Medium
SW-REQ-094	Storage temperature range handling	SYS-REQ-065	Medium
SW-REQ-095	Humidity constraint compliance	SYS-REQ-066	Low
SW-REQ-096	Continuous ADC acquisition	SYS-REQ-067	High
SW-REQ-097	ADC feedback control	SYS-REQ-068	High
SW-REQ-098	Hard stop at Shipping_ADC_Value	SYS-REQ-069	High
SW-REQ-099	UART ON/OFF control	SYS-REQ-070	High
SW-REQ-100	Fault condition detection	SYS-REQ-071	High
SW-REQ-101	Fault reporting	SYS-REQ-071	High
SW-REQ-102	State transition on fault	SYS-REQ-071	High
SW-REQ-103	State transition recovery	SYS-REQ-071	Medium
SW-REQ-104	State transition verification output	SYS-REQ-071	Medium
SW-REQ-105	Architecture diagram artifact generation	SYS-REQ-072	Low
SW-REQ-106	Schematic artifact generation	SYS-REQ-073	Low
SW-REQ-107	PCB layout artifact generation	SYS-REQ-074	Low
SW-REQ-108	Firmware source code packaging	SYS-REQ-075	High
SW-REQ-109	Calibration procedure documentation	SYS-REQ-076	Medium
SW-REQ-110	Test report generation	SYS-REQ-077	Medium
SW-REQ-111	Demonstration video packaging	SYS-REQ-078	Low

4. Requirements Summary

SW-REQ ID	Title	Category	Priority
SW-REQ-001	Closed-loop control algorithm	Functional	High
SW-REQ-002	ADC sampling for feedback	Performance	Medium
SW-REQ-003	UART command interface	Interface	High
SW-REQ-004	UART diagnostics reporting	Interface	Medium
SW-REQ-005	UART protocol settings	Interface	Medium
SW-REQ-006	Environmental condition handling	Constraint	Low
SW-REQ-007	Transient immunity robustness	Reliability	Medium
SW-REQ-008	EMC immunity robustness	Reliability	High
SW-REQ-009	MISRA-C compliance	Maintainability	Medium
SW-REQ-010	12V supply assumption	Constraint	Medium
SW-REQ-011	Potentiometer feedback configuration	Functional	High
SW-REQ-012	ADC read function	Functional	High
SW-REQ-013	ADC scaling to position	Functional	Medium
SW-REQ-014	Actuator control output	Functional	High
SW-REQ-015	Control loop timing	Performance	Medium
SW-REQ-016	Safe state on feedback fault	Reliability	High
SW-REQ-017	Shipping position threshold configuration	Functional	High
SW-REQ-018	Stop at shipping position	Functional	High
SW-REQ-019	UART ON command handling	Interface	High
SW-REQ-020	UART OFF command handling	Interface	High

SW-REQ-021	UART command acknowledgement	Interface	Medium
SW-REQ-022	Continuous ADC acquisition loop	Functional	High
SW-REQ-023	ADC resolution configuration	Performance	High
SW-REQ-024	ADC sampling rate	Performance	High
SW-REQ-025	ADC as primary feedback variable	Functional	High
SW-REQ-026	ADC-only flap control inputs	Functional	High
SW-REQ-027	Flap actuation range enforcement	Constraint	High
SW-REQ-028	Control decisions based on real-time ADC	Functional	High
SW-REQ-029	Prohibit open-loop control	Constraint	High
SW-REQ-030	ADC validity monitoring	Functional	High
SW-REQ-031	Permit motion only in ADC window	Constraint	High
SW-REQ-032	Reject out-of-window targets	Constraint	High
SW-REQ-033	Disable drive on ADC out of range	Functional	High
SW-REQ-034	Stop at target ADC	Functional	High
SW-REQ-035	Configure supply voltage ADC channel	Interface	High
SW-REQ-036	Compute scaled supply voltage	Functional	High
SW-REQ-037	Continuous supply monitoring	Performance	High
SW-REQ-038	Configurable overvoltage threshold parameter	Maintainability	Medium
SW-REQ-039	Overvoltage detection	Functional	High
SW-REQ-040	Disable drive on overvoltage	Functional	High
SW-REQ-041	Block actuator commands during overvoltage	Functional	High
SW-REQ-042	Enter fault state on overvoltage	Functional	High
SW-REQ-043	Overvoltage UART fault message	Interface	High

SW-REQ-044	Maintain FAULT state until clear conditions	Functional	High
SW-REQ-045	FAULT exit voltage check	Performance	Medium
SW-REQ-046	Continuous overvoltage monitoring	Functional	High
SW-REQ-047	UART actuator control interface	Interface	High
SW-REQ-048	Support ACTUATOR_ON command	Interface	High
SW-REQ-049	Support ACTUATOR_OFF command	Interface	High
SW-REQ-050	ACTUATOR_ON execution	Functional	High
SW-REQ-051	ACTUATOR_OFF execution	Functional	High
SW-REQ-052	Ignore invalid UART commands	Functional	Medium
SW-REQ-053	Log invalid UART commands	Functional	Medium
SW-REQ-054	Nominal voltage configuration	Constraint	Low
SW-REQ-055	Operating voltage range configuration	Constraint	High
SW-REQ-056	Safe outputs on startup	Reliability	High
SW-REQ-057	Startup delay for transient suppression	Reliability	Medium
SW-REQ-058	ADC resolution configuration	Functional	High
SW-REQ-059	ADC reference voltage configuration	Constraint	High
SW-REQ-060	ADC sampling frequency	Performance	High
SW-REQ-061	ADC overvoltage detection	Reliability	High
SW-REQ-062	ADC pin initialization	Interface	High
SW-REQ-063	Actuator GPIO initialization	Interface	High
SW-REQ-064	PWM timer initialization	Functional	High
SW-REQ-065	UART initialization	Interface	Medium
SW-REQ-066	DMA usage for data transfer	Performance	Medium

SW-REQ-067	Independent watchdog service	Reliability	High
SW-REQ-068	Control loop timing stability	Performance	High
SW-REQ-069	Layered architecture enforcement	Maintainability	Medium
SW-REQ-070	State machine implementation	Functional	High
SW-REQ-071	Initial state on reset	Functional	High
SW-REQ-072	Nominal state transitions	Functional	Medium
SW-REQ-073	Fault state transitions	Reliability	High
SW-REQ-074	Watchdog initialization	Reliability	High
SW-REQ-075	Watchdog service interval	Reliability	High
SW-REQ-076	Reset on detected malfunction	Reliability	High
SW-REQ-077	ADC-only feedback for control	Functional	High
SW-REQ-078	Control loop period	Performance	High
SW-REQ-079	Threshold-based actuator stop	Functional	High
SW-REQ-080	Anti-oscillation hysteresis	Performance	Medium
SW-REQ-081	Raw ADC usage only	Constraint	High
SW-REQ-082	Periodic UART status message	Interface	Medium
SW-REQ-083	UART baud rate configuration	Interface	High
SW-REQ-084	UART frame format configuration	Interface	High
SW-REQ-085	UART transmit interval	Performance	High
SW-REQ-086	Overcurrent shutdown	Functional	High
SW-REQ-087	Thermal shutdown	Functional	High
SW-REQ-088	Stall detection	Functional	High
SW-REQ-089	Short-circuit protection	Functional	High

SW-REQ-090	Undervoltage detection	Functional	High
SW-REQ-091	Watchdog supervision	Reliability	High
SW-REQ-092	Fault state transition	Functional	High
SW-REQ-093	Operating temperature range handling	Constraint	Medium
SW-REQ-094	Storage temperature range handling	Constraint	Medium
SW-REQ-095	Humidity constraint compliance	Constraint	Low
SW-REQ-096	Continuous ADC acquisition	Functional	High
SW-REQ-097	ADC feedback control	Functional	High
SW-REQ-098	Hard stop at Shipping_ADC_Value	Functional	High
SW-REQ-099	UART ON/OFF control	Interface	High
SW-REQ-100	Fault condition detection	Functional	High
SW-REQ-101	Fault reporting	Functional	High
SW-REQ-102	State transition on fault	Functional	High
SW-REQ-103	State transition recovery	Functional	Medium
SW-REQ-104	State transition verification output	Interface	Medium
SW-REQ-105	Architecture diagram artifact generation	Constraint	Low
SW-REQ-106	Schematic artifact generation	Constraint	Low
SW-REQ-107	PCB layout artifact generation	Constraint	Low
SW-REQ-108	Firmware source code packaging	Constraint	High
SW-REQ-109	Calibration procedure documentation	Constraint	Medium
SW-REQ-110	Test report generation	Constraint	Medium
SW-REQ-111	Demonstration video packaging	Constraint	Low